

Swaptions

7.1 The instrument

A **swaption** is an option to enter an interest rate swap at a future date. A **payer swaption** gives its holder the right, but not the obligation, to enter a payer swap (pay fixed, receive floating) at a predetermined strike rate K on the **expiry** date T ; a **receiver swaption** gives the right to enter a receive-fixed swap. The underlying swap, if exercised, runs for a further period called the **tenor**. A swaption is conventionally named by both numbers together — a “1y × 10y” swaption expires in one year into a ten-year swap.

Because entering a payer swap struck at K is worth $A(T) (S(T) - K)$ at the expiry date T (from Chapter 4’s swap valuation, evaluated at the point the swap begins), a rational holder exercises exactly when $S(T) > K$. The payer swaption payoff is therefore

Key result: the swaption payoff

$$\text{Payer swaption payoff at } T = N A(T) \max(S(T) - K, 0)$$

$$\text{Receiver swaption payoff at } T = N A(T) \max(K - S(T), 0)$$

where $S(T)$ is the forward swap rate observed at expiry and $A(T)$ is the annuity of the underlying swap, both as defined in Chapter 4. A payer swaption is a call on the swap rate; a receiver swaption is a put. The annuity plays exactly the role a single discount factor played for a caplet — it is what converts an option-on-a-rate payoff into money.

7.2 Pricing under the annuity measure

Exactly as the caplet price collapsed cleanly under the T_2 -forward measure because F_t is driftless there, the swaption price collapses cleanly under the **annuity measure** $\mathbb{Q}^A$ (numeraire $A(t)$), under which the forward swap rate $S(t)$ is a driftless martingale — again, proved properly in Chapter 12, used freely here. This gives

Key result: swaption pricing formula

$$\text{Swaption}(0) = N A(0) \mathbb{E}^{\mathbb{Q}^A} [\max(S(T) - K, 0)]$$

Assuming lognormal dynamics for $S(t)$ under $\mathbb{Q}^A$ gives **Black’s swaption formula**:

$$\text{Payer}_{\text{Black}}(0) = N A(0) [S \Phi(d_1) - K \Phi(d_2)], \quad d_{1,2} = \frac{\ln(S/K) \pm \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$$

Assuming normal dynamics gives **Bachelier’s swaption formula**:

$$\text{Payer}_{\text{Bachelier}}(0) = N A(0) \sigma_n \sqrt{T} [d \Phi(d) + \varphi(d)], \quad d = \frac{S - K}{\sigma_n \sqrt{T}}$$

identical in form to the caplet formulas of Chapter 6, with $F \rightarrow S$ and $P(0, T_2) \rightarrow A(0)$.

This structural parallel — swaption formulas are caplet formulas with the discount factor replaced by the annuity — is not a coincidence. Both are instances of the same general fact: an option on a martingale, priced under that martingale’s own natural measure, always reduces to a Black-or-Bachelier-type formula. Chapter 12 generalises this observation completely.

Worked example 7.1 — pricing the reference swaption.

Using the 3-year swap from Worked Example 4.1 ($S = 4.00\%$, $A(0) = 2.775$) reinterpreted as the underlying of a $1y \times 3y$ payer swaption struck at $K = 4.00\%$ (at the money), notional \$50mm, quoted at a normal volatility of $\sigma_n = 78$ bp/yr, expiry $T = 1$.

Step 1 — ATM simplification. Since $S = K$, $d = 0$, and $\Phi(0) = 0.5$, $\varphi(0) = 1/\sqrt{2\pi} = 0.39894$, so the bracket $d\Phi(d) + \varphi(d)$ collapses to $\varphi(0) = 0.39894$.

Step 2 — undiscounted ATM price in rate units.

$$C_n(S) = \sigma_n \sqrt{T/2\pi} = \sigma_n \sqrt{T} \times \varphi(0) = 0.0078 \times 1 \times 0.39894 = 0.003111$$

which is 31.11 bp of rate — the ATM Bachelier premium, expressed as a rate rather than money, needs only $\sigma_n \sqrt{T/2\pi}$.

Step 3 — convert to cash via the annuity and notional.

$$V = N \times A(0) \times C_n(S) = \$50,000,000 \times 2.775 \times 0.003111 \approx \$431,700$$

Sanity check. The ATM formula $\sigma_n \sqrt{T/2\pi}$ is a widely used mental-arithmetic shortcut precisely because $\sqrt{1/2\pi} \approx 0.399$ is close to 0.4: a trader hearing “78 normal on the 1y3y” can estimate the rate-unit premium as roughly $0.4 \times 78 \text{ bp} \approx 31 \text{ bp}$ without a calculator, then multiply by annuity and notional. This matches Step 2 exactly.

Interpretation. \$431,700 is the fair premium today for the right to enter a 3-year, \$50mm payer swap at 4.00% in one year’s time. Because the swaption was struck exactly at the current forward, its entire value is time value — there is no intrinsic component — which is why this number is driven purely by σ_n , T , the annuity, and the notional, with no dependence on $S - K$ at all.

7.3 Physical versus cash settlement

Two settlement conventions exist, and they price at slightly different levels because they use different annuities.

ON THE DESK

A **physically settled** swaption, on exercise, results in the actual underlying swap coming into existence, discounted using the real market discount curve for the annuity $A(T)$ — this is what the formula above assumes and is the dominant USD convention. A **cash-settled** swaption, common in EUR markets, instead pays a cash amount computed using a standardised, formula-based annuity built purely from the settlement swap rate itself (an “internal” discounting, ignoring the real curve’s shape) rather than the true market annuity. The pricing difference between the two is usually small but not zero, and a system that hard-codes one settlement convention while pricing instruments that trade under the other will show a small, persistent, and easily misdiagnosed pricing bias.

PITFALL

It is tempting to think of a swaption as “just a caplet on a swap rate instead of a LIBOR rate,” and structurally that is correct — but the swap rate $S(t)$ is a considerably more complicated stochastic object than a single forward rate $L(t; T_1, T_2)$. From the par-rate formula in Chapter 4, $S(t)$ is a *weighted average of many individual forward rates*, with weights that are themselves random (since they depend on discount factors). This means $S(t)$ ’s volatility and distribution are not primitive inputs to the BGM model — they are *derived*, in a generally non-trivial way, from the volatilities and correlations of the individual forward rates that make up the swap. Chapter 17 is devoted almost entirely to this one problem: given a BGM model calibrated at the level of individual forward rates, what swaption volatility does it imply, and can the model be tuned so that implied and market swaption vols agree?

Chapter benchmark

Before moving on, you should be able to, without notes:

1. Write the payer and receiver swaption payoffs in terms of the forward swap rate and annuity.
2. Price an at-the-money swaption using the Bachelier ATM shortcut $N A(0) \sigma_n \sqrt{T/2\pi}$, and explain why the ATM case has no intrinsic-value component.
3. Explain the structural difference between a caplet’s underlying (a single forward rate) and a swaption’s underlying (a weighted average of many forward rates), and why this makes swaptions harder to reconcile with a BGM model than caplets are.

